

■ Government Document Discovery & Crawling Guide

■ Overview

To build a repository of government documents, we need to index them. This is typically accomplished through web crawling, but additional discovery methods can enhance coverage and reach.

■ Primary Discovery Methods

1. Web Crawling – Using tools like Scrapy and Sitemaps
2. Search Engines – Google/Bing/Common Crawl APIs
3. RSS Feeds – Press releases and update feeds
4. Sitemaps & robots.txt – To map out structure
5. Dark Web Engines – Ahmia, Hidden Wiki for .onion docs
6. Wayback Machine – For historical or removed documents

■ Advanced Discovery Techniques

1. IP-Based Crawling – Use masscan/nmap to find web servers via IP
2. ASN/IP Range Mapping – Use BGP/IP tools to identify govt IPs
3. Certificate Transparency Logs – Find new domains from public TLS certs
4. DNS Brute Forcing – Tools like Amass and Subfinder
5. Ping + Banner Grabbing – Identify software and exposed services
6. Printer/FTP Server Scraping – Rare sources for exposed documents

■ Tools by Category

Web Crawling: Scrapy, Playwright, BeautifulSoup
Search/Feed Indexing: Google API, Common Crawl, Ahmia
Document Parsing: pdfminer.six, textract, pytesseract
IP Scanning: nmap, masscan, whatweb
Metadata Discovery: crt.sh, Censys, Amass
NER & NLP: spaCy, Hugging Face, LexNLP
Storage: PostgreSQL, Qdrant, Elasticsearch

■ Ethical & Legal Notes

- ✓ Public .gov/.mil crawling is generally safe
- IP scanning or deep scraping is jurisdiction-sensitive

■ Avoid unauthorized access to admin panels or sensitive data
Use .onion scraping responsibly and legally

■ Next Steps

1. Build a crawler for known .gov/.mil endpoints
2. Integrate IP scanning for shadow services
3. Use AI to classify, extract metadata, and summarize
4. Index into a vector database with search capabilities
5. Visualize relationships and trends across agencies/documents